CSE 471: MACHINE LEARNING

Learning Probabilistic Models

ML parameter learning (Continuous model)

$$P\left(x\right) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(x-\mu\right)^2}{2\sigma^2}}$$

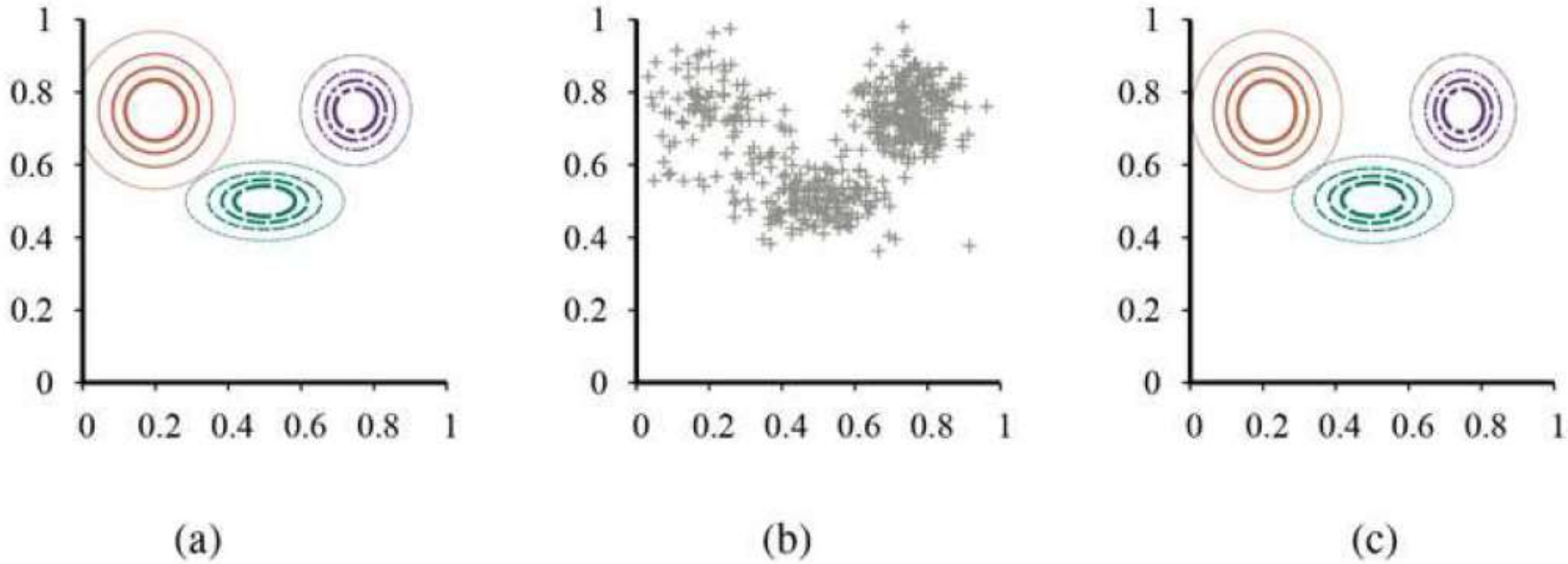
$$L = \sum_{j=1}^{N} \log \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(x_j-\mu\right)^2}{2\sigma^2}} = N\left(-\log\sqrt{2\pi} - \log\sigma\right) - \sum_{j=1}^{N} \frac{\left(x_j-\mu\right)^2}{2\sigma^2}$$

$$\frac{\partial L}{\partial \mu} = -\frac{1}{\sigma^2}\sum_{j=1}^{N}\left(x_j-\mu\right) = 0 \qquad \Rightarrow \qquad \mu = \frac{\sum_j x_j}{N}$$

$$\frac{\partial L}{\partial \sigma} = -\frac{N}{\sigma} + \frac{1}{\sigma^3}\sum_{j=1}^{N}\left(x_j-\mu\right)^2 = 0 \qquad \Rightarrow \qquad \sigma = \sqrt{\frac{\sum_j\left(x_j-\mu\right)^2}{N}}$$

The EM Algorithm

Gaussian mixture model



(a) A Gaussian mixture model with three components; the weights (left-to-right) are 0.2, 0.3, and 0.5. (b) 500 data points sampled from the model in (a). (c) The model reconstructed by EM from the data in (b).

Mixture distribution

- *k* components
- Each component is a distribution itself

$$P\left(\mathbf{x}\right) = \sum_{i=1}^{k} P\left(C = i\right) P\left(\mathbf{x} \middle| C = i\right)$$

- ***C is the random variable denoting the component***

Mixture of Gaussians

- Each component is a multivariate Gaussian
- Parameters
  - The weight of each component, $w_i = P(C = i)$
  - The mean of each component, (mu)
  - The covariance of each component $\sum_i$

Mixture of Gaussians

- If we knew the parameters of each component
  -> we could assign each data point to a component.

Gaussian Probability Density Formula

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Mixture of Gaussians

- If we knew which component generated each data point -> It would be easy to recover the component Gaussians

$$L = \sum_{j=1}^{N} \log \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(x_j-\mu\right)^2}{2\sigma^2}} = N\left(-\log\sqrt{2\pi} - \log\sigma\right) - \sum_{j=1}^{N} \frac{\left(x_j-\mu\right)^2}{2\sigma^2}$$

$$\frac{\partial L}{\partial \mu} = -\frac{1}{\sigma^2}\sum_{j=1}^{N}\left(x_j-\mu\right) = 0 \qquad \Rightarrow \qquad \mu = \frac{\sum_j x_j}{N}$$

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Solving by EM Algorithm

- Initialize the parameters arbitrarily (randomly initialized)
- E-Step

expectation step

pdf apply করবো।

E-step: Compute the probabilities $p_{ij} = P(C=i|\mathbf{x}_j)$, the probability that datum $\mathbf{x}_j$ was generated by component i. By Bayes' rule, we have $p_{ij} = \alpha P(\mathbf{x}_j|C=i)P(C=i)$. The term $P(\mathbf{x}_j|C=i)$ is just the probability at $\mathbf{x}_j$ of the ith Gaussian, and the term $P(C=i)$ is just the weight parameter for the ith Gaussian. Define $n_i = \sum_j p_{ij}$, the effective number of data points currently assigned to component i.

weight calculation এ কাজে লাগবে।
কোনো component এ কতগুলো datapoint
assign করলাম।

Solving by EM Algorithm

- Initialize the parameters arbitrarily
- M-Step

maximization step

M-step: Compute the new mean, covariance, and component weights using the following steps in sequence:

$$n_i = \sum_j p_{ij}$$

$$\mu_i \leftarrow \sum_j p_{ij}\mathbf{x}_j / n_i$$

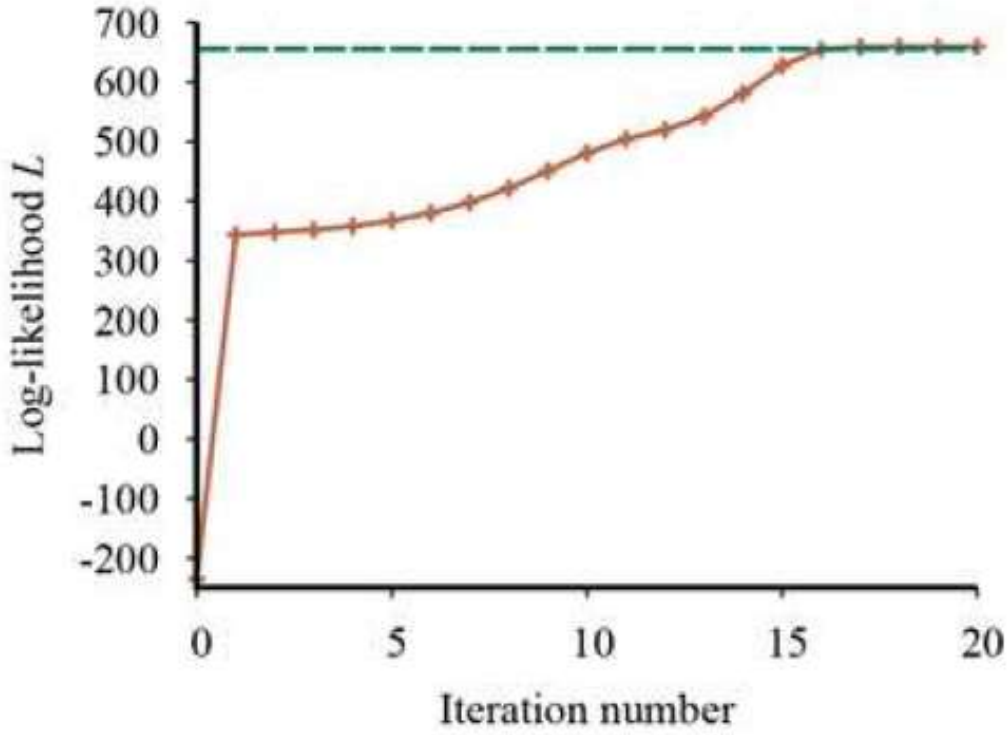
$$\Sigma_i \leftarrow \sum_j p_{ij}\left(\mathbf{x}_j - \mu_i\right)\left(\mathbf{x}_j - \mu_i\right)^T / n_i$$

component এর probability

$$w_i \leftarrow n_i / N$$

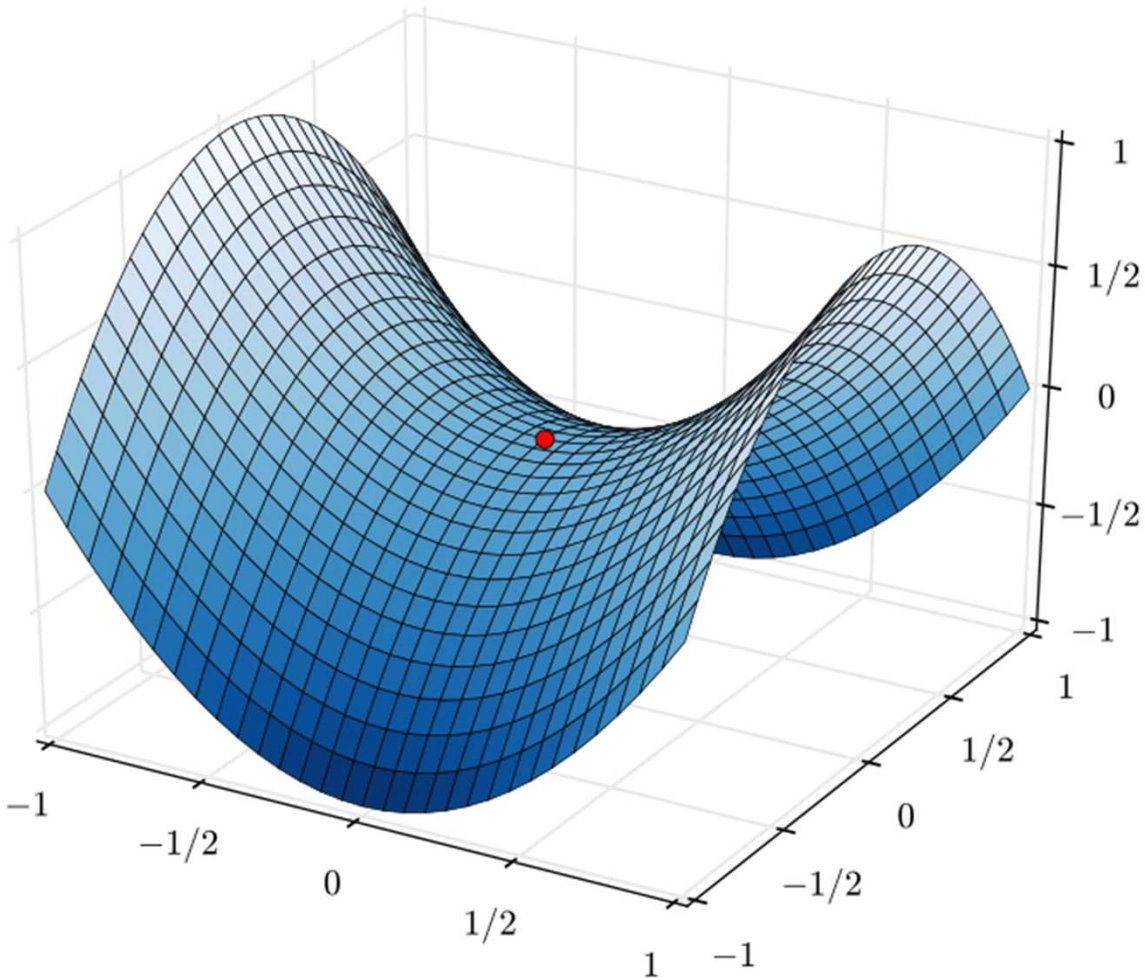
E-M loop এ চলবে।

Log likelihood increasing



Graphs showing the log likelihood of the data, L, as a function of the EM iteration. The horizontal line shows the log likelihood according to the true model. Graph for the Gaussian mixture model in Figure 20.12.

Saddle Point

→ condn মানলেও
guaranteed যে local minimum এ যাবে।
problem: stuck করতে পারে
→ saddle point

Soln: Random restart